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| <b>EXP NO : 7</b> | <b>BUILD A SIMPLE KNOWLEDGE<br/>BASE USING FIRST-ORDER LOGIC</b> |
| <b>DATE :</b>     |                                                                  |

**AIM:**

To build a simple knowledge base using First-Order Logic (FOL), store facts and rules, and process user queries to derive intelligent conclusions using Python.

**ALGORITHM:**

1. Import Libraries:

Import required modules from the kanren library: Relation, facts, run, var, and conde.

2. Define Relations:

Define FOL relations such as student, likes, and others.

3. Insert Facts into the Knowledge Base:

Use facts() to define who is a student and what subjects they like.

4. Define Rules:

Create a rule: "If a person is a student and likes something, then they are intelligent".

5. Accept User Query:

Take user input (e.g., "Who is intelligent?" or "Is Alice intelligent?")

6. Perform Reasoning:

Based on the query, use run() and the defined rules to determine the correct answer.

7. Display Result:

Print whether the person is intelligent or list all intelligent people

## PROGRAM:

```
import collections
import collections.abc
collections.Iterator = collections.abc.Iterator
collections.Hashable = collections.abc.Hashable
from kanren import Relation, facts, run, var, conde

doctor = Relation()
treats = Relation()

facts(doctor, ("dr_smith",), ("dr_jones",),
      ("dr_lee",), ("dr_williams",), ("dr_davis",))

facts(treats, ("dr_smith", "flu"),
      ("dr_smith", "allergy"), ("dr_jones", "cold"),
      ("dr_lee", "covid"), ("dr_lee", "pneumonia"),
      ("dr_williams", "diabetes"), ("dr_davis", "hypertension"))

def is_skilled(person):
    disease = var()
    return conde((doctor(person), treats(person, disease)))

query = input("Enter Query: ").lower().strip()
x = var()

if query.startswith("who"):
    result = run(0, x, is_skilled(x))
    if result:
        print("✓ Skilled doctors found:")
        for name in result:
            print(f"- {name.replace('_', ' ').title()}")
    else:
        print("X No skilled doctors found.")
```

```
elif query.startswith("is"):
    words = query.replace("?", "").split()

    if len(words) >= 3:
        name = words[1]
        result = run(0, x, is_skilled(x))

        if name in result:
            print(f"✓ Yes, {name.replace('_', ' ').title()} is skilled.")
        else:
            print(f"X No, {name.replace('_', ' ').title()} is not skilled.")

    else:
        print("X Invalid query format.")
```

## OUTPUT:

```
PS D:\GangaProjects\AIA> python Exp7.py
Enter Query: is dr_davis skilled ?
✓ Yes, Dr Davis is skilled.
PS D:\GangaProjects\AIA> python Exp7.py
Enter Query: is dr_ganga skilled ?
X No, Dr Ganga is not skilled.
PS D:\GangaProjects\AIA> python Exp7.py
Enter Query: who are skilled doctors?
✓ Skilled doctors found:
- Dr Davis
- Dr Smith
- Dr Jones
- Dr Lee
- Dr Williams
```

## RESULT:

The First-Order Logic (FOL) knowledge base program was executed successfully. It correctly processed the given facts and rules, answered user queries, and verified the output as per the defined intelligent person rule.